



Multi-UAV Collaborative Face Recognition for Goods Receiver in Edge-Based Smart Delivery Services

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Abstract. The accurate and efficient recognition of goods receivers is a crucial task for Unmanned Aerial Vehicle (UAV) based smart delivery services in edge computing. Meanwhile, face recognition has become one of the major recognition methods in smart delivery services given its high efficiency and reliability. However, unlike many other application scenarios, the face recognition algorithm's accuracy and efficiency are often limited due to the unique camera shooting angles by the UAVs flying in the sky. To address such an issue, in this paper, we propose a multi-UAV collaborative face recognition method to enhance both the accuracy and efficiency of face recognition for multiple goods receivers at the same time, named UAVs4FR. Specifically, instead of each UAV running its own face recognition process, multiple UAVs first use face detection models to collaboratively detect and capture pedestrian faces, then transmit these images to the edge server. Afterwards, the edge server uses a face recognition algorithm to perform preliminary recognition of faces. Meanwhile, the face fusion model is also deployed in the edge server to fuse the face images to improve the recognition accuracy. Finally, the edge server utilizes a face recognition algorithm to further recognize target goods receivers using the fused face images and send out the corresponding instructions to the group of UAVs through broadcasting. Comprehensive experiments based on a real-world edge-based smart UAV delivery service system successfully demonstrate the effectiveness of our proposed framework and its better performance compared with other representative methods.

Keywords: Edge Computing · UAV · UAV Smart Logistics · Multi-UAV Collaboration · Face Recognition

1 Introduction

Smart logistics is a modern logistics model supported by modern information technology to achieve intelligent control and improve operational efficiency in all stages of cargo transportation and distribution [1, 2]. In recent years, indepth integration of Internet of Things (IoT) and Artificial Intelligence (AI) in the logistics industry has further promoted the development of smart logistics, making it gradually become an effective solution for improving the quality and efficiency in the logistics industry [3–5]. As an important intelligent delivery device, Unmanned Aerial Vehicles (UAVs) have broad application prospects in many fields such as smart logistics and modern agriculture because of their flexibility, low costs, and ease of deployment [6, 7]. Due to its important role in the “last mile”, such as epidemic sealing and control and complex terrain areas, UAV delivery has gradually been widely used in various smart logistics scenarios [8, 9]. For example, ecommerce companies such as JD and Amazon have started establishing a UAV delivery system that uses UAVs for goods delivery [10, 11]. However, traditional cloud-based UAV smart logistics cannot guarantee real-time and efficient services due to factors such as network instability and large data transmission [12]. Edge computing-based UAV smart logistics has been proposed to address these issues by performing data computation closer to the edge nodes [13, 14].

Despite the many advancements in edge computing, there are still many open technical challenges deploying UAVs for smart logistics. One of the key challenges is how we can accurately and efficiently identify goods receivers with face recognition technologies [15]. However, many of the current face recognition algorithms only have high recognition accuracy for frontal faces, and in actual scenarios, the recognition accuracy will be significantly reduced due to several factors such as variable face angles, which is a key challenge when applied to UAV smart logistics [16].

There are several studies on improving face recognition accuracy in the context of UAV scenarios. For example, Shao et al. [18] use facial detection algorithms to detect pedestrians’ faces and then used face recognition networks to identify them. However, in practical applications, most of the faces captured by UAVs have random postures, and this method cannot guarantee high accuracy in face recognition. Amato et al. [19] designed a specialized network structure for face recognition in low-resolution conditions for UAVs. This method ensures face recognition in low-quality facial images. However, this method ignores the negative impact of facial poses. Xu et al. [20] used a generative adversarial network to train a face frontalization model and then used this model to frontalize faces. This method can eliminate the negative impact of facial angle on face recognition to some extent, and can improve face recognition accuracy. However, the effectiveness of face frontalization model varies for different faces, resulting in some instability in practical scenarios. Shen et al. [21] proposed a facial scoring mechanism and UAV flight control algorithm that adjusts the flight direction according to the captured facial pose, enabling the UAV to automatically capture frontal facial photos for face recognition. Because this method ensures captur-

ing the front face of a pedestrian every time, this method significantly increases face recognition accuracy in smart logistics for UAVs because it ensures capturing the front face of the pedestrians. However, UAV trajectory adjustments require considerable time, hence reducing the efficiency of identifying the target recipient. Abate et al. [22] proposed a three-dimensional facial reconstruction method based on UAVs, where the pedestrian first attracts the UAV's attention using gestures. The UAV then hovers around the pedestrian to collect sufficient face information to construct a 3D facial model, which is used for face recognition. Although this method constructs a 3D facial model with good distinctive features, recognizing pedestrian gestures and constructing 3D faces is time-consuming, significantly impacting face recognition efficiency. Gao et al. [23] first used posture recognition to filter pedestrians and then used frontal facial capture to perform face recognition. This method also significantly improves face recognition accuracy. However, the posture recognition process requires high hardware requirements, which cannot be guaranteed without appropriate hardware capabilities, thus affecting the efficiency of identifying the target goods receiver. In this paper, we propose a face recognition method based on multi-UAV collaboration to improve the recognition accuracy of face recognition algorithms in UAV smart logistics scenarios. Firstly, the UAVs use face detection models to detect and capture pedestrian faces, then transmits them to the edge server. Secondly, the edge server uses a face recognition algorithm to perform preliminary recognition of faces. Thirdly, the face fusion model deployed in the edge server is used to fuse the faces of pedestrians, which results in fused faces. Finally, the edge server utilizes a face recognition algorithm to further recognize the fused face. If successful, confirmation instructions are sent to the UAVs through broadcast and the subsequent commodity delivery task is accomplished. Our experimental results using real-world scenarios and datasets demonstrate that the proposed framework further improves the efficiency of face recognition while ensuring high recognition accuracy of the face recognition algorithms.

2 Framework Design and Workflow

In this section, we provide an overview of our proposed framework and explore in detail the working details of each module in it.

2.1 Framework Overview

In this section, we will give an overall introduction to our proposed framework, named UAVs4FR. As shown in Fig. 1, the framework is divided into three main modules: (i) cloud server module, (ii) edge server module, and (iii) UAV group module as a terminal device. The cloud server module is responsible for storing information such as frontal photos and delivery addresses uploaded by users according to system prompts during registration, and continuously provide intelligent services for the UAV smart logistics system. The edge server has sufficient computing resources and is closer to the terminal device compared to the cloud

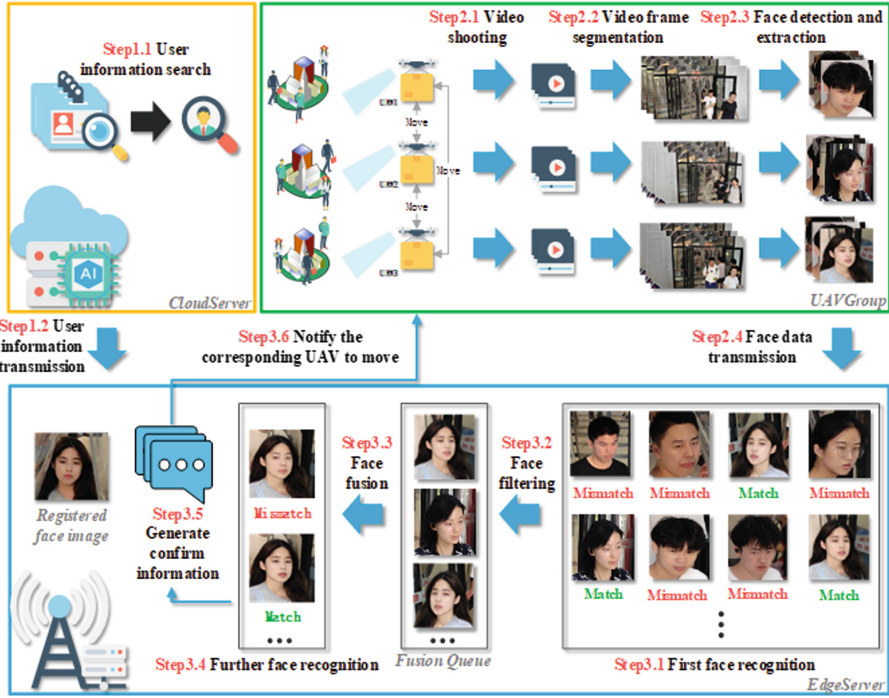


Fig. 1. The architecture of UAVs 4FR

server, which can effectively reduce the data transmission time during the service. Therefore, AI models and algorithms are required for the service, such as face fusion models and face recognition algorithms, which can be deployed in the edge server. As terminal devices in the delivery system, the UAVs use their equipped hardware devices (e.g., cameras, batteries) to perform flight and commodity delivery-related tasks. In addition, the UAVs can run some lightweight applications such as LFFD [24] or tiny-yolo [25] to perform data pre-processing tasks (such as face detection or face interception) by using limited computational resources.

2.2 Framework Execution Process

We now describe the workflow of our proposed framework in detail through a practical example. The relevant definitions are described in Table 1. The specific process can be divided into the following steps:

Step 1: CS processing user data.

Step 1.1: CS search P according to order information.

Step 1.2: CS transfer and temporarily store P into ES.

Step 2: U processing of captured pedestrian videos.

Step 2.1: U uses the equipped high-definition camera to take pictures of pedestrians.

Table 1. Relevant definitions

Symbol	Meaning
k	Number of users with the same receiving location
w	Number of pedestrian faces
CS	Cloud Server
ES	Edge Server
$R = \{\gamma_1, \gamma_2 \dots \gamma_k\}$	The set of goods receivers
$P = \{p_1, p_2 \dots p_k\}$	The set of registered faces corresponding to goods receivers
$U = \{u_1, u_2 \dots u_k\}$	Group of UAVs distributed by the system for goods receivers
$F = \{f_1, f_2 \dots f_w\}$	The set of pedestrian faces
α	The threshold for first face recognition
β	The threshold for further face recognition
$Q = \{q_1, q_2 \dots q_k\}$	The threshold for further face recognition
$C = \{c_1, c_2 \dots c_k\}$	The threshold for further face recognition
t	Time for face fusion queue
$L = \{l_1, l_2 \dots l_k\}$	The set of confirmation instructions
v	Face feature value

Step 2.2: U uses the limited on-board resources to split the captured pedestrian video into video frames and retain some of the video frame data at a specific ratio.

Step 2.3: The lightweight face detection model deployed in U performs face detection on the retained video frames, and the detected faces are cropped out using an image segmentation algorithm, each cropped face is bound with pedestrian location information and the corresponding UAV information, these face images constitute F.

Step 2.4: U transmits F to ES.

Step 3: ES performs target goods receiver identification tasks according to F

Step 3.1: The face recognition model deployed in s matches the faces in F with the registered faces in P in turn, and the matching result is compared with α . If the matching result is less than or equal to α , the matching is successful, otherwise the matching fails.

Step 3.2: ES filter the faces in F, the specific steps are as follows: discard the faces that do not match with all registered faces in P, deposit f_j that matches successfully with p_i into the corresponding q_i , and set $c_i = t$ for q_i , which can continue to deposit other pedestrian faces before $c_i \leq 0$.

Step 3.3: If $c_i \leq 0$, the face fusion model deployed in ES fuses the faces in q_i in turn in order to obtain more discriminative faces.

Step 3.4: The face recognition model deployed in s matches the fused faces with p_i , and the matching result is compared with β . If it is less than or equal to β , the matching is successful, otherwise the matching fails.

Step 3.5: For the fused face successfully matched with p_i , ES generates l_i , and at the same time obtains the position of u_i , calculates the landing direction of u_i based on the pedestrian positioning information bound in the face and the current position of u_i , and stores it in l_i Step 3.6: ES sends l_i to U with broadcast.

Step 4: u_i adjusts the direction of landing according to the information stored in l_i and completes subsequent delivery tasks.

3 Related Technologies and Algorithms

In this section, we describe the relevant technologies and the algorithms used in the framework.

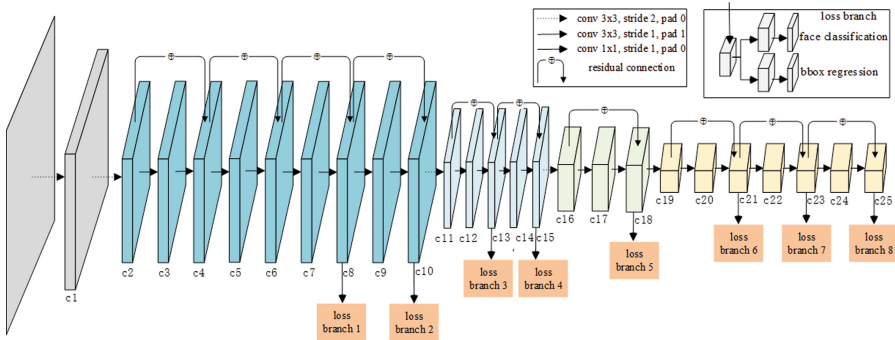


Fig. 2. The architecture of LFFD

3.1 Face Detection

Face detection technology is a critical aspect of automatic face recognition systems, which uses certain strategies and algorithms to determine whether a face is present in a given image. In our proposed architecture, the face detection task is performed by the lightweight face detection model LFFD deployed in the UAV, which is suitable for application to UAVs with limited on-board computational resources because of its small size and fast detection. The framework of LFFD is shown in Fig. 2. As shown in Fig. 2, LFFD has 25 convolution layers divided into four parts for various Receptive Field (RF) strategies. There are also eight loss branches which are used to detect faces with different scales. Each branch has

two sub-branches for facial classification and bounding box regression. LFFD adopts softmax with cross-entropy loss over two classes for facial classification. For bounding box regression, it uses L2 loss directly. The regression ground truth is defined as follows:

$$\frac{RF_x - b_x^{tl}}{RF_s/2}, \frac{RF_y - b_y^{tl}}{RF_s/2}, \frac{RF_x - b_x^{br}}{RF_s/2}, \frac{RF_y - b_y^{br}}{RF_s/2} \quad (1)$$

where RF_x and RF_y are the center coordinates of the RF, RF_s is the RF size and $RF_s/2$ is the normalization constant, b_x^{tl} and b_y^{tl} are coordinates of the top-left corner of the bounding box, b_x^{br} and b_y^{br} are coordinates of the bottom-right corner of the bounding box. Based on this method, faces with different scales in video frames will be predicted by the RF of a certain convolution layer and then extracted accordingly.

3.2 Face Fusion

With the continuous improvement and development of AI technology, the image processing technology using face fusion is being applied to many different scenarios. Face fusion refers to the process of fusing two face pictures into one face picture, and the face obtained through face fusion will have the features of the faces in both pictures [26]. At this stage, face fusion technology has a wide range of applications in image processing, video editing, and other fields. Face fusion is mainly divided into the following steps: First, face key point extraction, through which 68 key points specific to the face will be obtained; Second, the triangulation of the face region based on the face key points; Third, an affine transformation of the face triangle region; Fourth, set the transparency for face fusion, with the fusion formula shown below:

Algorithm 1. Face Fusion Process

Input: f_a, f_b

Output: fused face f_c

- 1: Extract face feature points for f_a and get the set of face key points kp_a
 - 2: Extract face feature points for f_b and get the set of face key points kp_b
 - 3: Delaunay triangulation of f_a according to kp_a to obtain the corresponding set of triangles T_a
 - 4: Delaunay triangulation of f_b according to kp_b to obtain the corresponding set of triangles T_b
 - 5: **for** each $i \in [1, 68]$ **do**
 - 6: Calculate the mean value T_c^i of T_a^i and T_b^i
 - 7: Affine transform T_a^i and adjust its position to coincide with T_c^i
 - 8: **end for**
 - 9: Perform face fusion based on (2) to obtain f_c
 - 10: **return** f_c
-

$$M(x, y) = \alpha \cdot f_a(x, y) + (1 - \alpha) \cdot f_b(x, y) \quad (2)$$

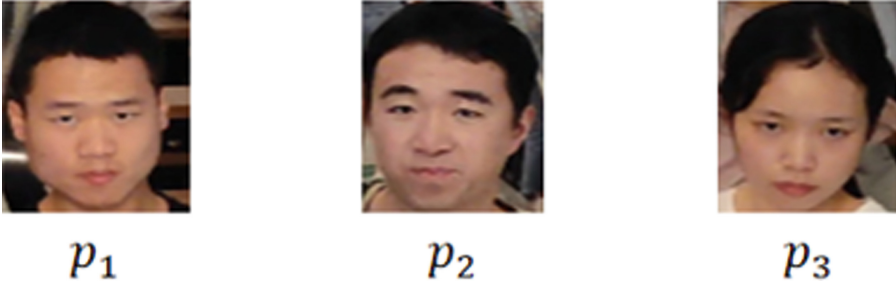


Fig. 3. Users' registered face images

where $M(x, y)$ denotes the pixel value of the face image after fusion, α is the fusion degree parameter, $f_a(x, y)$ denotes the pixel value of the face image f_a with coordinates (x, y) to be fused, and $f_b(x, y)$ denotes the pixel value of the face image f_b with coordinates (x, y) to be fused. The face fusion code is process is presented in Algorithm 1.

3.3 Face Recognition

Face recognition is a biometric identification technology based on human facial feature information, which provides the pre-requisites for identification due to the uniqueness of the human face and the difficulties in replication. In the face recognition process, the Euclidean distance (ED) between face feature values is usually used as the judgment criterion, and the function is defined as shown in (3).

$$ED(F_i, F_j) = \sum_{n=1}^{128} \sqrt{(x_{in} - x_{jn})^2} \quad (3)$$

The values of face features F_i and F_j comprise 128 feature vectors x_{in} and x_{jn} respectively (where $n = 1, 2, \dots, 128$). The Euclidean Distance (ED) is calculated between F_i and F_j . If the ED result is less than the pre-defined threshold value, then it is concluded that the two face features correspond to the same person, otherwise, they are not considered to be the same [27]. In the context of UAV smart logistics, the face recognition system based on deep learning can achieve satisfactory results under ideal acquisition conditions and user cooperation. However, the face photos captured by image devices are often in random poses, which adversely affects the accuracy face recognition algorithms. To address this challenge, we used face fusion technology that combines the faces captured by multiple UAVs to improve the accuracy of face recognition during the goods receiver identification process. The specific process is presented in Algorithms 2 and 3.

Algorithm 2 instructs the edge server to execute varying operations depending on the results of face recognition. In cases where pedestrian faces exhibit a high level of similarity, the edge server immediately produces related confirmation information. However, for pedestrian faces with lower similarity, the decision to

Algorithm 2. Face Recognition Process**Input:** P, F, α, β and t **Output:** Q, L

```

1: Initialize  $Q = \{q_1, q_2 \dots q_k\}$ 
2: Initialize  $C = \{c_1, c_2 \dots c_k\}$ 
3: for each  $i \in [1, w]$  do
4:   Calculate  $v_i$  corresponding to  $f_i$ 
5:   for each  $j \in [1, k]$  do
6:     if  $c_j > 0$  then
7:       Calculate  $v_i$  corresponding to  $p_j$ 
8:       Calculate ED between  $v_i$  and  $v_j$  according to (3)
9:       if  $ED \leq \beta$  then
10:         $l_i \leftarrow \text{ture}$ 
11:         $c_j = 0$ 
12:       else
13:        if  $\beta \leq ED$  and  $ED \leq \alpha$  then
14:          if  $q_j = \text{null}$  then
15:             $q_j \leftarrow f_i$ 
16:             $c_j = t$  and start the countdown
17:          else
18:            if  $c_j > 0$  then
19:               $q_j \leftarrow f_i$ 
20:            end if
21:          end if
22:        end if
23:      end if
24:    end if
25:  end for
26: end for
27: return  $Q, L$ 

```

add them to the corresponding face fusion queue is made based on time. For pedestrians in the face fusion queue, we used fusion techniques to create more distinctive faces. Algorithm 3 elaborates on this process.

4 Experiments and Evaluation

4.1 Case Studies

This section provides a practical example of how the proposed framework operates in identifying a goods receiver. Three users have placed orders with delivery addresses. The framework queries them as $p = \{p_1, p_2, p_3\}$. The details are shown in Fig. 3. Subsequently, the system assigns one UAV to each goods receiver for efficient delivery of the goods. A UAV group, denoted by $U = \{u_1, u_2 \dots u_k\}$. Additionally, ES located close to the delivery addresses is responsible for downloading P from CS and temporarily storing it. When U arrives at the destination, it captures footage of the surrounding pedestrians using the configured camera.

Algorithm 3. Face Fusion-Based Goods Receiver Recognition**Input:** P, Q, β **Output:** L

```

1: for each  $i \in [1, k]$  do
2:   if  $l_i = \text{null}$  then
3:     Calculate the number of elements in  $q_j$  and note it as  $N$ 
4:     if  $N \leq 1$  then
5:        $l_i \leftarrow \text{false}$ 
6:     else
7:        $q_i^1 = f_a$ 
8:       for each  $j \in [2, N]$  do
9:          $q_i^j = f_b$ 
10:        Fuse  $f_a, f_b$  into  $f_c$  according to Algorithm (1)
11:        Calculate  $v_c$  of  $f_c$ 
12:        Calculate  $v_i$  corresponding to  $p_i$ 
13:        Calculate ED between  $v_c$  and  $v_i$  according to (3)
14:        if  $ED \leq \beta$  then
15:           $l_i \leftarrow \text{true}$ 
16:        else
17:          Select the image more similar to  $p_i$  from  $f_a$  and  $f_b$  as  $f_a$ 
18:        end if
19:      end for
20:      if if all elements in  $q_i$  are fused and  $l_i = \text{null}$  then
21:         $l_i \leftarrow \text{false}$ 
22:      end if
23:    end if
24:  end if
25: end for
26: return  $L$ 

```

The recorded video is split into frames at a 30:1 ratio and the resulting partial video frames are displayed in Fig. 4. The face detection algorithm LFFD deployed in U is capable of detecting faces in the video frames and capturing the intercepted faces. A set of face images obtained from this interception is denoted as $F = \{f_1, f_2 \dots f_w\}$. These images are subsequently transmitted by U to ES. A few of the captured face images are displayed in Fig. 5. ES calculates ED between the face in F and each registered face in P using the face recognition algorithm. If the ED calculated by f_7 and p_1 satisfies $ED \leq \beta$, then ES sets $c_1 = 1$. Additionally, ES calculates the landing direction of u_1 based on the positioning information bounded by f_7 and the location of u_1 at that moment. The landing direction is stored in l_1 , and ES then sends l_1 to U in the form of a broadcast. Once u_1 receives l_1 , it proceeds to perform the commodity delivery task. Subsequently, f_9 and p_2 calculate ED to satisfy $\beta \leq ED \leq \alpha$. ES then makes $c_2 = 30$ and begins counting down, storing f_9 in q_2 . Similarly, f_{12} and q_3 calculate ED satisfy $\beta \leq ED \leq \alpha$. ES makes $c_3 = 30$ and starts counting down, storing f_{12} in q_3 . Suppose q_2 and q_3 each deposit four faces before $c_2 = 0$ and $c_3 = 0$, the faces deposited by q_2 are f_9, f_{13}, f_{17} , and f_{20} , while the faces deposited by q_3

are f_{12} , f_{15} , f_{18} , and f_{27} . The faces stored in q_2 and q_3 are illustrated in Fig. 6. The fused facial features are shown in Fig. 7, where q_2^1 is the face obtained by fusing f_9 with f_{13} , q_2^2 is the face obtained by fusing f_{13} with f_{17} , q_2^3 is the face obtained by fusing f_{13} with f_{20} , q_3^1 is the face obtained by fusing f_{12} with f_{15} , and q_3^2 is the face obtained by fusing f_{15} with f_{18} . During further facial recognition, neither q_2^1 and q_2^2 were successfully matched, while q_2^3 was matched with p_2 . Based on the positional information of f_{20} and the current location of u_2 , ES calculated the landing direction of u_2 and stored it in l_2 . ES then broadcasts l_2 to U. q_3^1 did not match with p_3 , while q_3^2 matched successfully with p_3 . In this case, ES terminates the face fusion and face recognition process in advance, then calculated the landing direction of u_3 based on the positional information of f_{18} and the current location of u_3 , and stored it in l_3 . ES subsequently broadcasts l_3 to U, and u_3 received l_3 to complete the delivery task.



Fig. 4. Video frames

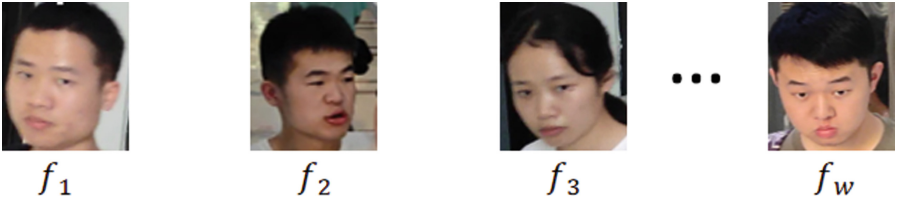


Fig. 5. Pedestrians' faces

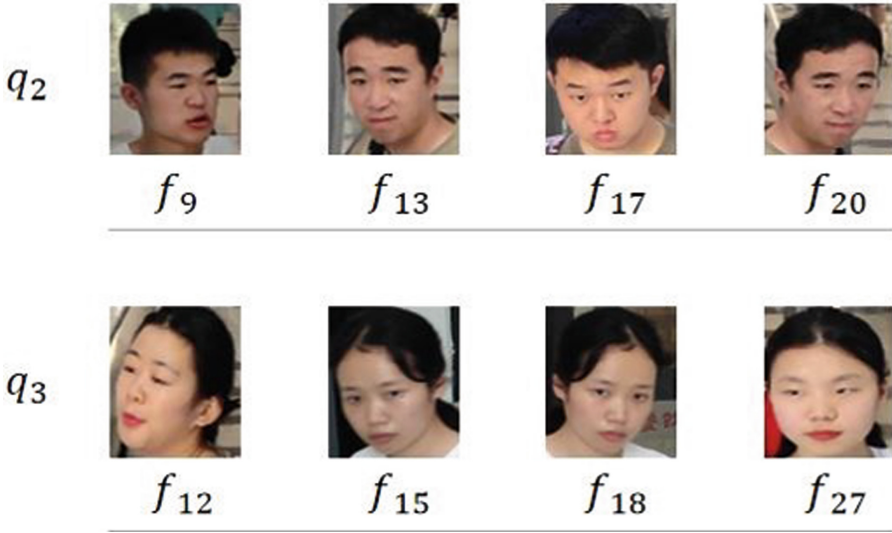


Fig. 6. Faces in q_2 and q_3

4.2 Evaluation

In this section, we compare the accuracy and efficiency of our proposed framework with several other representative device-edge collaboration frameworks to evaluate the effectiveness of the proposed architecture. The representative architectures are shown as follows:

- **Baseline:** After receiving the face image transmitted by the UAV, the edge server performs direct face recognition.
- **Edge4FR:** The edge server initially receives the face image uploaded from the UAV and then processes it using a face frontalization model to generate a corresponding frontal face image. It then conducts face recognition tasks on the frontal face image.
- **Edge4Sys:** The edge server first receives face images transmitted by the UAV and performs pose recognition to filter out pedestrians that meet recognition conditions. Eligible pedestrians' frontal faces are then captured and further processed through face recognition.
- **PC4FR:** The private cloud server first receives the face images uploaded by the UAVs and processes them through a face recognition algorithm to identify eligible face images. Eligible face images then undergo face fusion operations using the face fusion model deployed in the private cloud server. Finally, fused face images undergo further face recognition through the face recognition algorithm. The dataset used in the experiments was collected by a DJI Matrice 600 Pro UAV and consists of images of approximately 150 people. All experiments were conducted on computers running Windows 10

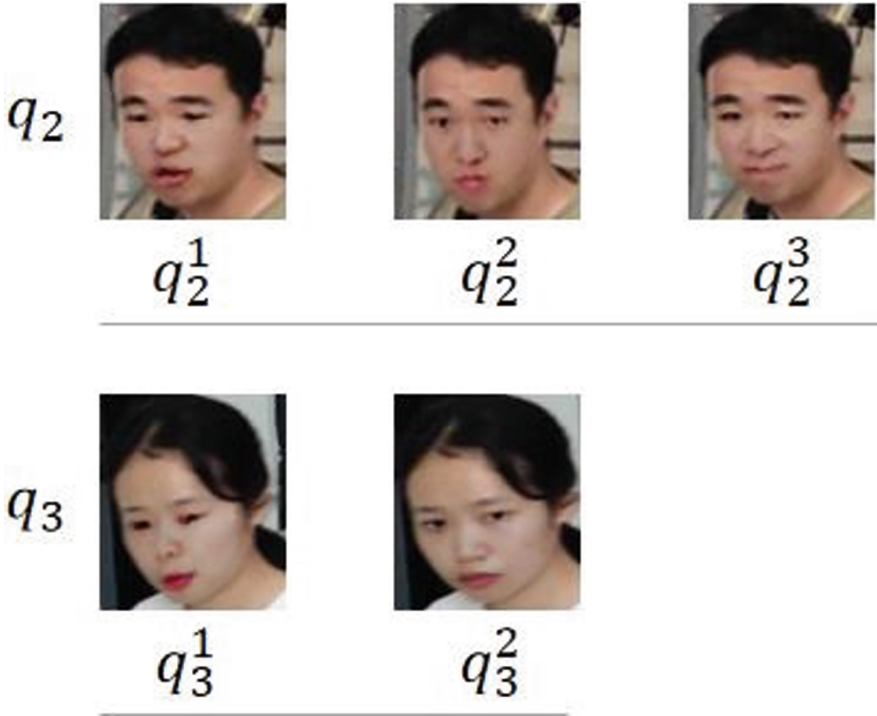


Fig. 7. Fused faces

(64-bit) operating system, using an 11th generation Intel® Core™ i5-11400 processor and 16 GB of RAM.

Accuracy. The accuracy comparison results for the four architectures are shown in Fig. 8. The Baseline framework has the lowest face recognition accuracy since it cannot effectively address potential issues caused by random UAV captures such as random face poses. The Edge4FR framework improves face recognition accuracy to some extent. However, it should be noted that face frontalization through Generative Adversarial Network (GAN) training. Due to the training data or the network itself, the face frontalization model may not have a positive effect on every face image, which limits the improvement of face recognition accuracy. As demonstrated in our experimental results, the Edge4Sys framework obtains the highest recognition accuracy. However, the process of locating the goods receiver using pose recognition is time-consuming and unreliable. The UAVs4FR framework prescreens face images more similar to the goods receiver through preliminary face recognition before performing more accurate face recognition. It then performs face fusion operations on the screened face images to obtain face images with more distinctive features. Although its recognition accu-

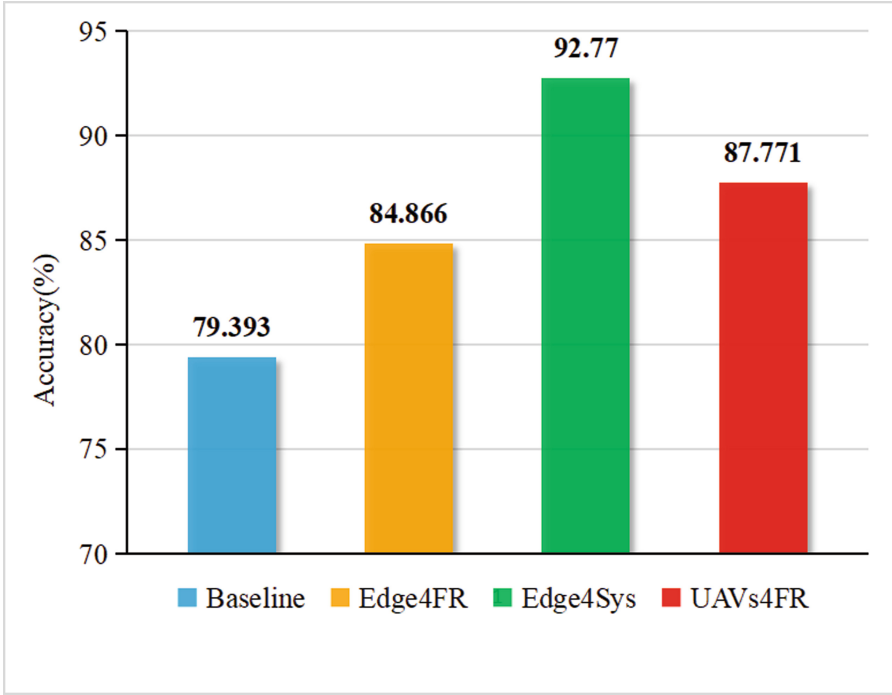


Fig. 8. Comparison of identification accuracy

racy is slightly lower than the Edge4Sys framework, it is significantly higher than the Baseline framework and Edge4FR framework.

Efficiency. We now compare our proposed framework with several other frameworks in terms of face recognition efficiency. The time taken to recognize the goods receiver under different frameworks is illustrated in Fig. 9. Specifically, blue color denotes the time used for face detection and face extraction in the Baseline, Edge4FR, and UAVs4FR frameworks (FP). Purple color denotes the time used for face recognition in the Baseline, Edge4FR, and Edge4Sys framework (FR). Yellow color denotes the time taken for face turnaround in the Edge4FR (FF), and orange is the time consumed for pedestrian detection and pedestrian extraction in the Edge4Sys framework (PP). Green color is the time used for pose recognition in the Edge4Sys framework (PR), and red color indicates the time used for face recognition and face fusion in the UAVs4FR framework (TFR). We can observe that the Edge4Sys framework spends a considerable amount of time in the pose recognition process, making its overall time the highest among the four frameworks. Furthermore, as the number of UAVs in the scene increases, the time taken by the other three frameworks to complete the goods receiver recognition task increases significantly. Conversely, the UAVs4FR framework has a slower over-all time growth and takes significantly less time than the other

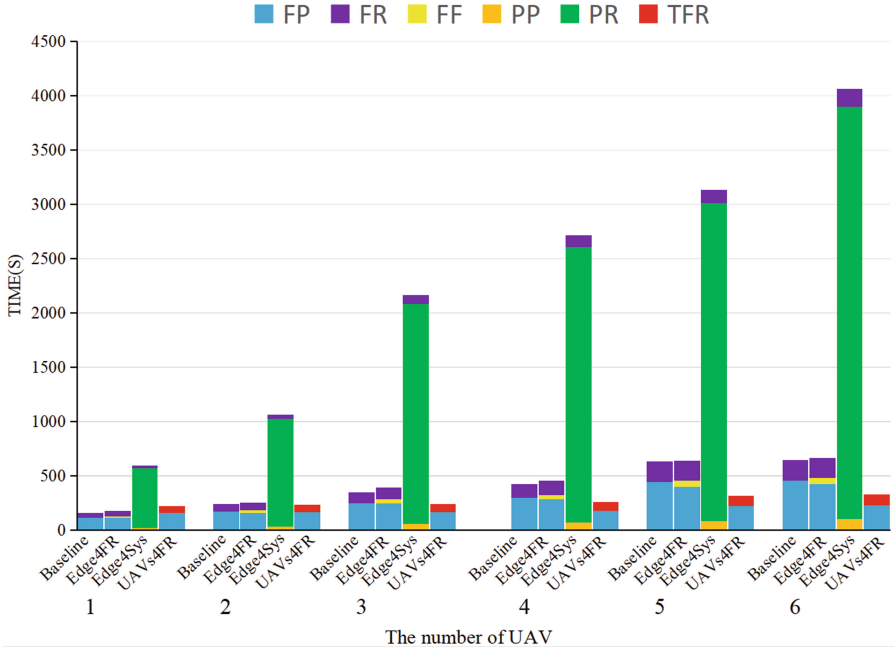


Fig. 9. Time comparison of identifying the goods receiver through different frameworks (Color figure online)

three frameworks, thanks to its multiUAV collaboration scheme. The results of the time comparison between UAVs using collaborative and non-collaborative approaches to complete goods receiver identification are presented in Fig. 10. We can observe that as the number of UAVs increases, the time taken to complete goods receiver identification in a noncollaborative manner increases significantly whereas the time taken in a collaborative approach by multiple UAVs increases slowly. This highlights the effectiveness of the collaborative UAV solution proposed in our architecture as it can significantly reduce the time required to complete goods receiver identification, thereby improving the overall efficiency of UAV-enabled smart logistics delivery. To further demonstrate the importance of the edge server in the proposed framework for face recognition, we compare the efficiency of our proposed framework with Private4FR. The experimental results are illustrated in Fig. 11, with different colors indicating different stages of the process. The orange color represents the time taken for face detection and extraction in the PC4FR framework ($PC_F P$). The yellow color represents the time for data transfer between the PC4FR framework and the end device ($PC_D T$). The green color represents the time for completing face fusion and recognition in the PC4FR framework ($PC_T F R$). The blue color represents the time taken for face detection and extraction in the UAVs4FR framework ($UAV_s F P$). The red color represents the time taken for face fusion and recognition in the UAVs4FR

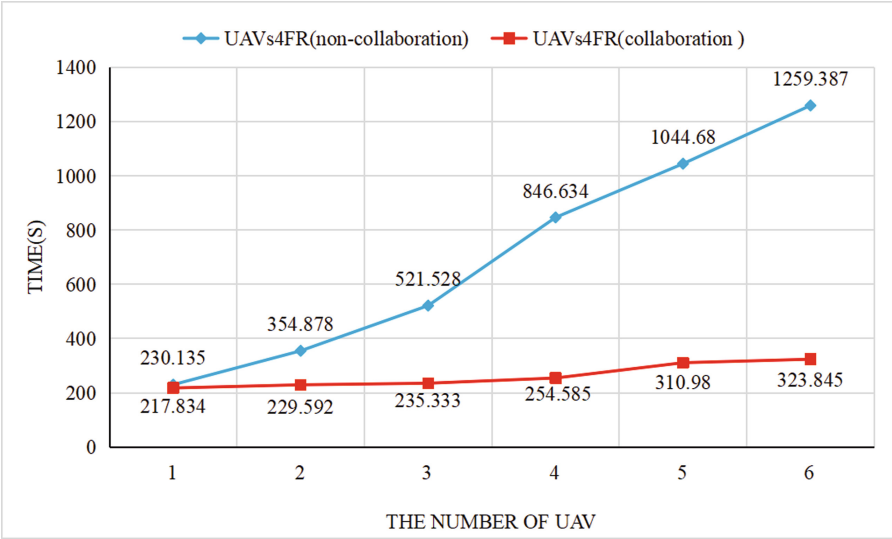


Fig. 10. Time comparison of identifying the goods receiver through collaborative and non-collaborative approach

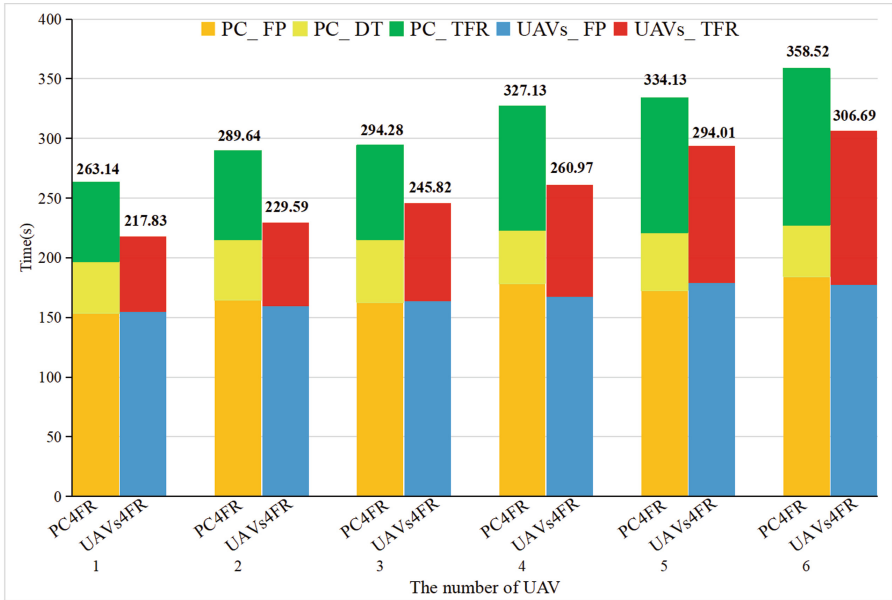


Fig. 11. Time comparison of identifying the goods receiver through Private4FR and UAVs4FR (Color figure online)

framework (UAV_{s_TFR}). The results clearly show that the time taken for face recognition in the UAVs4FR framework is significantly less compared to the private cloud server-based architecture. In the private cloud server-based architecture, the terminal devices upload all the captured and extracted face images to the private cloud server for recognition and fusion. This leads to a substantial amount of facial data, which can cause delays in data transmission between the cloud server and terminal devices.

5 Conclusions

In this paper, focusing on the goods receiver recognition task in the edge-based smart UAV delivery service system, we proposed a multi-UAV collaborative face recognition framework called UAVs4FR to address both the accuracy and efficiency issues in multiple goods receiver recognition. Our experimental results based on a real-world edge-based smart UAV delivery service scenario and datasets have successfully demonstrated the effectiveness of our proposed framework. There are still some potential future research directions in this area. For example, we only considered the involvement of one edge server in our proposed framework. The collaboration between multiple UAVs and multiple edge servers could be much more complex and will also require multiple levels of collaboration. We will explore this research problem as part of future work to further improve the performance of our proposed framework.

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